ICLR 2025 Reproduction

RuAG

Learned-Rule-Augmented Generation

for Large Language Models

Document-Level Relation Extraction · DWIE Dataset
MCTS Rule Discovery · Knowledge Injection · Mistral Small 24B

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Workflow: How to Implement

From paper to code — four stages

01 Data Infrastructure

Paper §4.1

- DWIE dataset: 802 docs, 23,130 entities
- SQLite database (ruag.db) with 7 tables
- Tables: documents, entities, relations,
- predictions, rules, relation_types, filtered_docs
- Code: data_loader.py

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02 Vanilla Baseline

Paper §4.3 + Fig. A1

- Figure A1 prompt: task desc, 20 relation defs,
- entity list, document content
- Temp = 0, top-p = 1 (Table A5)
- Strict exact-match validation against entity list
- Tested: Mistral Small, Code: baseline_vanilla.py

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03 MCTS Rule Discovery

Paper §3.1–3.2

- Search space filtered by entity co-occurrence
- UCT formula: $Q/N + C \cdot \sqrt{(\ln N_parent / N)}$
- 4 phases: Select → Expand → Simulate → Backprop
- C=0.7, 100 rollouts, max rule length = 2
- Code: mcts.py, rule_node.py, rule_evaluator.py

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04 Knowledge Injection

Paper §3.3 + Fig. A1

- 24 rules after cleaning (precision ≤ 0.5 removed)
- Rules translated to natural language via rules2text.py
- Injected into prompt at {rules} placeholder
- Model derives additional triples via rules
- Code: baseline_ruag.py

Issue Encountered & Solution

Free API Rate Limits

Problem

Mistral: 2 req/min. Groq: 500K tokens/day.
93 documents frequently trigger 429 errors.

Solution

Retry mechanism in call_llm(): on 429,
wait 15s, retry up to 5x. SQLite commits
after each doc — no data loss on crash.

Prompt Adaptation for Smaller Model

Problem

Paper's prompt targets GPT-4. Mistral outputs
"Bush" instead of "George W. Bush", misses
-x variants, mixes direct/rule-derived triples.

Solution

Added CRITICAL RULES (exact entity names,
-x variant + gpe0 explanation) and complete
worked example (5 direct + 1 rule-derived).

MCTS Search Space Explosion

Problem

64 relation types × combinatorial body
predicates = infeasible without filtering.

Solution

Entity co-occurrence filter from §3.1:
64 → 9–45 candidates per target.
Removed 3 irrelevant predicates (§4.1).

Results: Does Our Reproduction Match?

Model	Vanilla F1	RuAG F1	Improvement
Paper: GPT-3.5	18.94%	26.63%	+7.69%
Ours: Mistral Small 24B	**43.33%**	**51.82%**	**+8.49%**
Paper: GPT-4	46.94%	60.42%	+13.48%

Our +8.49% sits between GPT-3.5 (+7.69%) and GPT-4 (+13.48%)
— consistent with Mistral's intermediate capability.

Key Findings

Precision-Recall Tradeoff

-8.43% Prec **+16.52%** Rec

Top Rule Impact

head_of_state → citizen_of
(96.89% precision) Recall: 13.81% → **57.62%**

Absolute Gap

51.82% vs **60.42%** (GPT-4)
Gap due to model size difference

Per-Relation: Rules Help Most Where Vanilla Fails

Relation	Vanilla F1	RuAG F1	Change	Corresponding Rule
agent_of	12.41%	**52.60%**	**+40.19%**	head_of_state → agent_of (98.76%)
citizen_of-x	18.25%	**54.64%**	**+36.39%**	head_of_gov-x → citizen_of-x (97.98%)
citizen_of	23.58%	**59.46%**	**+35.88%**	head_of_state → citizen_of (96.89%)

Takeaways

1 Search Constraints Are Essential

Without entity co-occurrence filtering (§3.1), MCTS is intractable. A critical but easy-to-miss detail from the appendix.

2 Model Size > Prompt Engineering

Llama 8B → Mistral 24B (3x bigger): F1 jumped from 26% to 43%. Larger models also benefit 2.5x more from rules (+3.46% vs +8.49%).

3 SQLite Simplifies Everything

Single database for data, predictions, and rules. Made MCTS evaluation, checkpointing, and final scoring all query the same source.

4 Logic Rules Are Model-Agnostic

Our +8.49% validates the paper's core claim: MCTS rules consistently boost performance across GPT-3.5, GPT-4, and Mistral.

Verdict: Successfully validated RuAG — Search + Generation reliably improves Relation Extraction.